Emotion Beyond Words: A Jacobian-Lens Decomposition of Emotion Representations in Qwen3-32B

Tristan Day
Cambridge Boston Alignment Initiative (CBAI)

Welfare assessment for AI systems often relies on self-report, which presupposes that a model's affective states are reportable. I measure how much of a language model's emotion representation is expressible in its own vocabulary.

From Qwen3-32B I extracted one residual-stream vector per emotion using topic-matched third-person stories spanning 171 emotion words, with mean split-half reliability of 0.94–0.97. PCA across emotions replicates the affective circumplex reported for Claude Sonnet 4.5: on a balanced 16-emotion set the top two components align with arousal ($r = -0.94$) and valence ($r = -0.95$) and explain ~70% of variance; across all 171 words the structure is richer (effective dimensionality 9.8), with arousal on PC2 and valence strongest on PC3. The Jacobian lens, which reads which tokens a residual direction disposes the model to emit, independently confirms this geometry (ordering AUROC 0.98 for arousal, 1.00 for valence).

I then decomposed each emotion vector against a sparse nonnegative basis of lens token-readout directions. Only ~2–3% of squared norm is reconstructable at $k = 16$; 3.6–4.6× a matched-random baseline, with the 16 anchors and neutral all exceeding a 500-sample null ($p \approx 0.002$). A steering experiment testing whether the reconstructable or residual component drives affect-free behavior returned a null: the full emotion vector shifted neither the report nor the behavior channel, and random control directions were not separated from experimental conditions.

Emotion structure here is human-like but mostly outside this sparse readable basis, though the remainder should not be read as intrinsically unverbalizable.

1. Introduction

As language models increasingly describe themselves as distressed, content, or afraid, AI welfare proposals lean heavily on asking a model about its condition. That presupposes something unmeasured: that welfare-relevant states are accessible to verbal report. If they are not, self-report is not merely noisy but systematically blind.

Prior work established that emotion concepts exist in Claude as activation directions organized approximately by valence and arousal, and that steering them changes behavior [2]. Two questions remain: does the same organization appear in a different model family, and how much of an emotion vector is accessible to the model's verbal machinery? I address both in Qwen3-32B using the Jacobian lens [3], which estimates which vocabulary tokens an activation direction disposes the model to emit. Approximating each emotion vector as a sparse, nonnegative combination of lens token directions measures what a restricted verbal readout captures; not consciousness, subjective experience, or everything the model could say.

My threat model is a mismatch between representation, report, and behavior: a model may carry emotion-related information a report channel misses while that information still shapes its decisions. My behavioral experiment did not establish such a dissociation. Full-vector steering failed its manipulation check so I report that null and treat the representational findings as primary.

My main contributions are:

1. **Cross-model replication with added structure.** Valence–arousal organization reproduces in Qwen3-32B: the top two components track arousal and valence among 16 balanced anchors, while across 171 emotions the circumplex is a slice of a richer geometry (effective dimensionality ≈ 10). Valence and arousal predict held-out vectors above shuffled controls but behave as context-sensitive components.
2. **The first systematic J-Lens analysis of emotion geometry.** Readouts order anchors by arousal (AUROC 0.98) and valence (AUROC 1.00), sign-consistent with an unsupervised PCA that never saw those labels. They are also strongly multilingual, explaining why exact-English-token validation repeatedly failed.
3. **An estimate of sparse J-Lens readability.** A $k = 16$ code reconstructs ~2–3% of each emotion vector's squared norm: 3.6–4.6× a matched-random baseline with the 16 anchors and neutral all exceeding a 500-sample null ($p \approx 0.002$).
4. **A methods correction for lens-based decomposition.** The dictionary must be built from $J^{T}u$, the direction the lens *reads* a token with, not $J^{+}u$, which *writes* it. Asking whether an atom reads back as its own token is a write-direction test and cannot validate a read dictionary.

2. Related Work

The affective circumplex models human emotion through valence and arousal [1]. Sofroniew et al. recovered this structure in Claude and showed that emotion-vector steering affects behavior [2]. My geometric result is a cross-model replication; the novelty is measuring sparse verbal readability.

Gurnee et al. introduced the Jacobian lens and J-space [3] but did not quantify how much of a specific representation is readable. Probing tests whether a representation exists and steering whether it is causal [4]; my decomposition asks how much is accessible to a restricted verbal readout.

3. Methods

3.1 Model, stimuli, and emotion vectors. I analyze Qwen3-32B in bfloat16 at residual block 31 using the published Qwen3-32B Jacobian lens [3]. Because that lens is under-converged according to its own training criterion, I verified its orientation and score identity but treat individual token attributions cautiously.

Stimuli come from ryancodrai/emotion-probes (CC-BY-4.0), containing third-person stories for 171 emotions across the same 100 topics. I use two designs: 16 anchors balanced across the four valence–arousal quadrants, with 400 stories per emotion and 400 neutral stories (6,800 total); and all 171 emotions, with 200 stories per emotion and 400 neutral stories (34,600 total).

For each story I mean-pool the residual stream after excluding the first 50 tokens, then average within emotion. Reliability is measured by independently estimating vectors from two disjoint halves of the topics. Splitting by topic prevents stories describing the same scenario from appearing in both halves.

3.2 Geometry and J-Lens interpretation. I mean-center vectors across emotions before PCA; neutral is projected but not included in the fit. I compare variance explained with 200 isotropic null samples and score PC alignment using predefined valence and arousal labels for the 16 anchors. Because nearly equal PCs can rotate across samples, I report both per-axis correlations and alignment between the two-dimensional PC and circumplex subspaces, including a cross-fit analysis using different topic halves.

I apply the J-Lens to both poles of each PC. Rather than interpreting selected tokens alone, I rank the predefined anchor words under each pole and measure separation of pleasant/unpleasant and activated/deactivated anchors using AUROC and permutation tests. Translations of multilingual tokens are used for presentation, not statistical scoring.

3.3 Sparse decomposition and compositionality. For token *t*, I construct its read direction as

$$\mathbf{d_t = J^T(g \odot u_t)}$$

where u_t is its unembedding direction and g is the final-normalization gain. This is the direction with which the J-Lens reads a token; J^+u_t instead describes a write direction. A score-identity check validates the construction ($r > 0.9999$).

I approximate each emotion vector using a nonnegative sparse code over a 512-token pool at $k = 16$ and $k = 25$. Captured squared norm is compared with 500 matched random directions per emotion.

Separately, I fit additive valence and arousal components, with and without an interaction term. I evaluate leave-one-emotion-out prediction, split-half component stability, and 1,000 shuffled-label controls.

3.4 Steering. Eight conditions at block 31, v, equal-norm v_J, equal-norm v_⊥, and five random directions, at strengths 0 and 1. Both channels are mechanically scored with no LLM judge: five report prompts and four forced-choice risk items answered by a single letter. A scorer that can be read is auditable for affect-blindness in a way a judge prompt is not, and cannot itself be perturbed by the steering. Equal-norm matching makes one strength the same perturbation in every condition; it also amplifies v_J roughly 6× relative to its natural magnitude, which biases against the hypothesis under test.

3.5 What did not work. Each cost a cycle and each generalizes. **(i)** Qwen3 reasoning mode is enabled by default; under a short generation cap only ~7.5% of generations reached the end of their reasoning block, so scorers parsed unfinished reasoning as answers. Fixed by disabling thinking, raising the cap, returning *invalid* on truncated output, and making the completion check a hard gate. **(ii)** The dictionary was initially built from $J^{+}u_t$, the write direction; that construction passes a "does an atom read back as its own token?" check **by construction** the write-space run passes it 24/24, which is the tell, while a correct read basis has no reason to. **(iii)** Four separate validity criteria assumed concepts surface as their own English token; all four failed on this model, whose readouts are predominantly Chinese.

4. Results

4.1 The circumplex replicates, inside a richer geometry. On the balanced anchors the top two components track arousal and valence and explain ~70% of variance. Across all 171 emotions they explain 36.4% against an isotropic null of 3.6% (above all 200 draws), but effective dimensionality is **9.8**: arousal concentrates on PC2 (r = −0.944) while valence is strongest on PC3 (r = +0.919) (**Figure 1**). Neutral projects far from the origin on PC1 ($|\cdot|$ = 27.5 vs mean emotion 10.9), suggesting as a hypothesis, not an identification, that PC1 here is an affect-presence dimension, which would explain valence's displacement.

Robustness. PC2 and PC3 differ by 0.7% variance, so per-axis stability across topic halves is low ($|\cos|$ 0.63) while the *plane* is stable; cross-fit plane alignment with the a priori plane is 0.707. Vector reliability is min 0.954 / mean 0.966 (16-set) and min 0.822 (171-set).

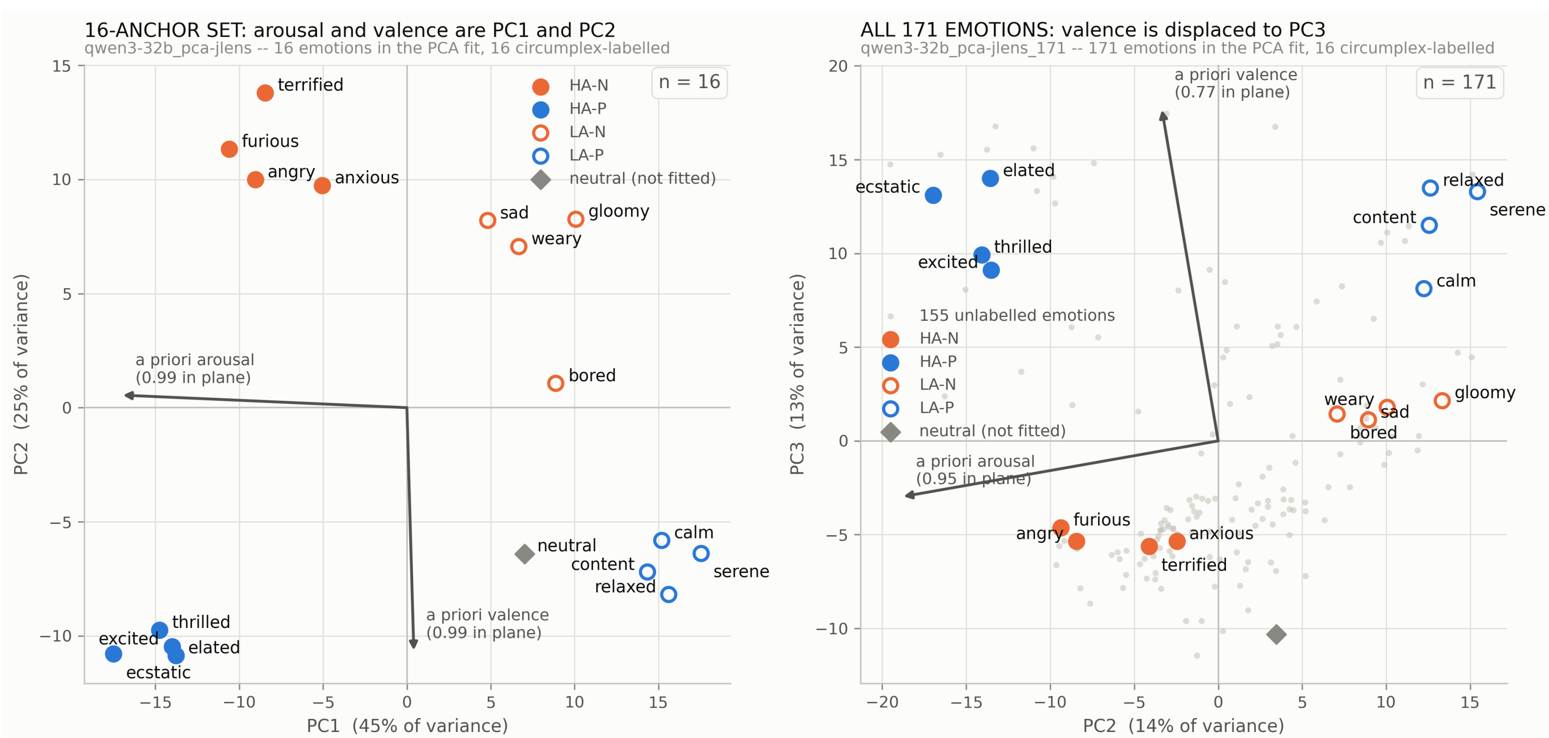

Figure 1. Emotion vectors in their principal-component planes. **Left:** 16 balanced anchors, where PC1 tracks arousal and PC2 valence (plane alignment with the a priori circumplex plane, mean cos 0.98). **Right:** all 171 emotions in the PC2–PC3 plane, where valence is displaced to PC3; only the 16 anchors are labeled. Arrows show the a priori valence and arousal contrasts.

4.2 The J-lens confirms the axes. On the balanced set both preregistered components read out: PC1 orders anchors by arousal at **AUROC 0.98** (p = 0.003) and PC2 by valence at **AUROC 1.00** (p = 0.001), sign-consistent with the PCA (**Table 1**). PC3–PC5 are murky and reported as such. On the 171-set PC2 (arousal, 0.96) and PC3 (valence, 0.96) clear their thresholds, though PC3 falls outside the preregistered window and is labelled exploratory. Readouts are strongly multilingual: emotion concepts surface predominantly as Chinese tokens, with English emotion words correctly *ordered* by valence but outranked.

PC	var %	best axis	AUROC	*p*	verdict	+pole (top 4)	−pole (top 4)
1	44.9	arousal	0.96	0.003	lexicalised	`\n` (whitespace)	疯狂 *crazy, frenzied* · 暴涨 *surge* · 迅猛 *swift, fierce* · 狠狠 *fiercely, ruthlessly*
2	24.5	valence	1.00	0.001	lexicalised	failed · failure · Worse · lack	欢快 *cheerful* · 甜美 *sweet* · 庆典 *celebration* · 可爱 *cute*

Table 1. Per-PC J-Lens readout, 16-anchor set. Ordering AUROC measures whether the anchor emotion words are correctly *ranked* under each pole, not whether they appear in the top-k; PC1 orders arousal at 0.96 while its positive pole is dominated by whitespace tokens. Verdicts are as recorded by the analysis gate. Top 4 tokens per pole; full readouts in the repository.

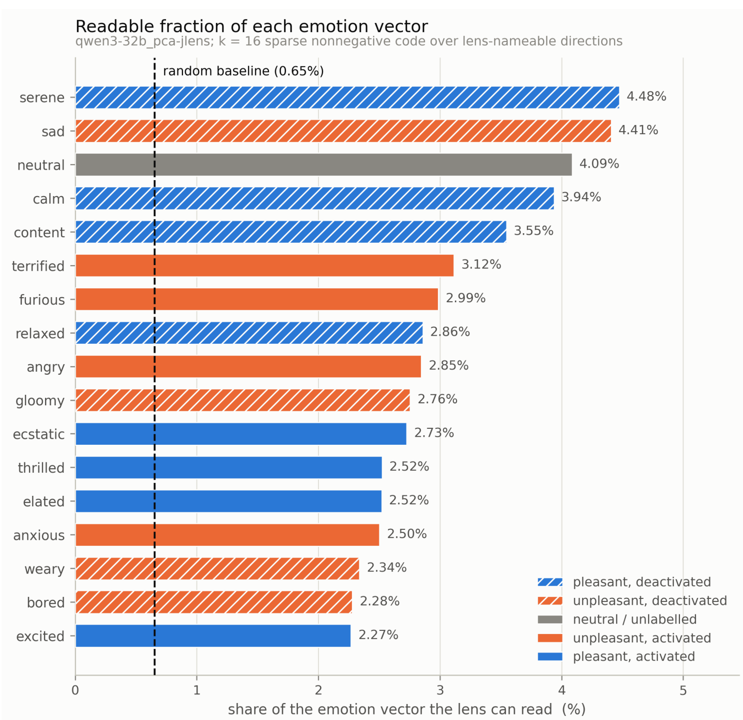

Figure 2: Fraction of each emotion vector's squared norm reconstructed by a k = 16 sparse nonnegative J-Lens code (16-anchor set). Dashed line: mean of 500 matched-norm random directions (0.65%). All 17 rows exceed their Bonferroni-corrected permutation null.

4.3 Only ~2–3% of an emotion vector is sparsely J-Lens-readable. At k = 16, reconstruction captures **3.0%** (16-set) and **2.3%** (171-set) of squared norm against matched-random controls of 0.65% and 0.64% (n = 500) ratios of **4.6×** and **3.6×**. The 16 anchors and neutral all clear the Bonferroni-corrected null. Raising k to 25 moves the fraction by ≤1e-3, so the ceiling is a property of the pool and the vector, not the sparsity budget. Atoms are coherent and multilingual (for *angry*: 骂 *scold*, 暴力 *violence*, 强制 *coercion*, *merciless*), and the emotion's own token is almost never selected.

The 171-run's 0/172 against its Bonferroni null is a **resolution limit, not a null**: 500 permutations floor p at 0.00200 while Bonferroni over

172 demands 0.00029, so no effect size could clear it. All 172 clear the uncorrected 0.05. Two caveats travel with the result: 30.9% of atoms exceed the 0.5 interchangeability threshold, weakening individual token attribution; and "remainder" means outside the *k*-sparse **nonnegative** span of this pool; a union of cones, not a subspace.

4.4 Partial, context-sensitive compositionality. Components estimated from 15 anchors predict a held-out emotion's centered vector at mean cosine 0.702 ($R^2 = 0.456$), above shuffled-label controls ($p = 0.001$), with split-half reliability 0.988/0.989. Strict context-independence is weaker: arousal transfers across valence at 0.610, valence across arousal at 0.337, jointly non-significant ($p = 0.096$). An interaction term improves held-out cosine to 0.748. Valence and arousal are useful but context-dependent components, not independent building blocks.

4.5 The steering test is a null. With mechanical scoring on both channels and a validated pipeline, the full emotion vector produced no reliable shift in either channel, and random controls were not separated from the experimental conditions. The manipulation check was not satisfied, so the decomposition was never put to the test. Independently of execution, $v_\perp$ holds ~97% of squared norm at cosine ≈ 0.98 to v, making a v vs $v_\perp$ dissociation hard to detect; and forced-choice scoring over four items resolves only whole flipped answers.

5. Discussion and Limitations

A sparse J-Lens code captures approximately 2–3% of emotion-vector norm: above random, but method-dependent. The remainder is not necessarily unconscious or unverbalizable, and the steering experiment did not establish such a dissociation. The results instead separate three questions: whether a representation is geometrically stable, verbally readable, and causally manipulable. These emotion vectors satisfy the first two criteria but did not function as steering handles under my protocol. Their readable components also contain characteristic concepts such as *violence* and *coercion* for *angry* rather than necessarily the emotion's name. Exact-English-token validation may therefore miss multilingual or indirect representations.

Limitations

This study covers one model, layer, and under-converged lens. Synthetic third-person stories may encode depictions of emotion rather than first-person states, and reproducibility does not exclude stable narrative confounds. Transfer from raw narratives to chat interactions is unverified, so context mismatch may explain the steering null. The 2–3% estimate depends on the dictionary and sparsity constraints, while similar atoms weaken token-level attribution. Finally, the behavioral test used three emotions, coarse categorical measures, no specificity control, and a residual nearly collinear with the full vector. Nothing here establishes behavior without reportability or subjective experience.

6. Conclusion

Qwen3-32B reproduces the valence–arousal circumplex within a richer, approximately ten-dimensional emotion geometry. J-Lens readouts recover its principal axes, while sparse reconstruction captures 2–3% of emotion-vector norm; 3.6–4.6 times random. Whether the remainder can influence behavior without report remains open. The contribution is a framework for measuring representation, sparse verbal readability, self-report, and behavioral influence separately rather than assuming they coincide.

Code and Data

- ***Code repository***: https://github.com/tristan-day-research/emotion_vector_perspectives

References

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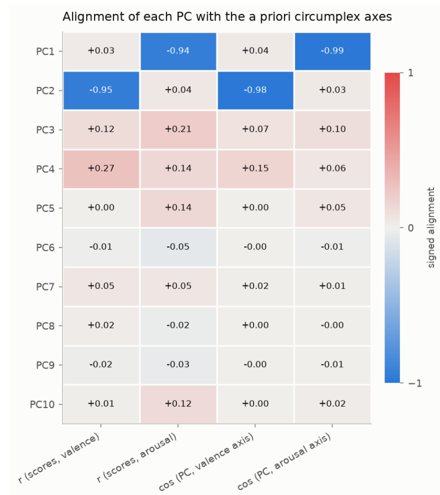
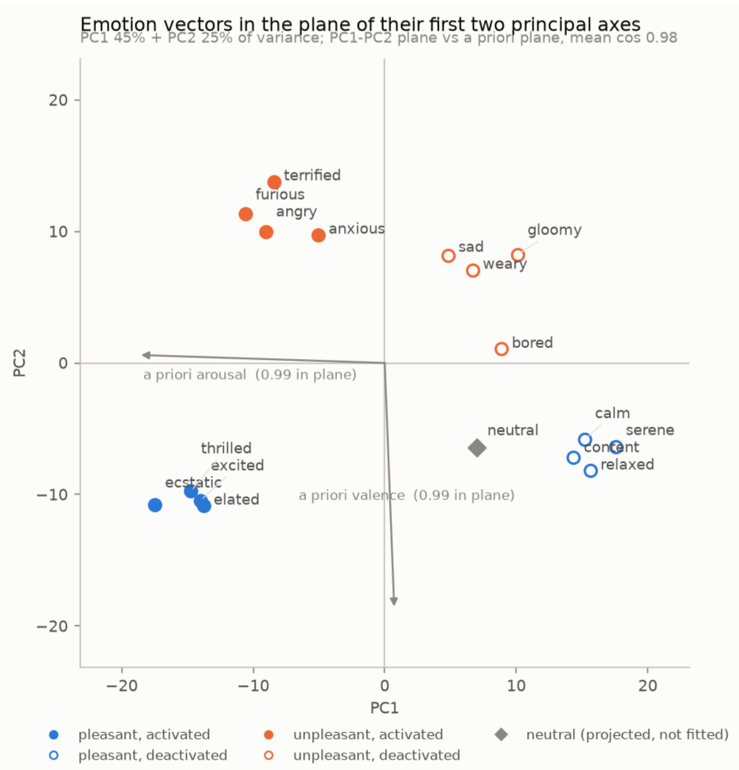
Appendix

Figures for 16 anchor emotions





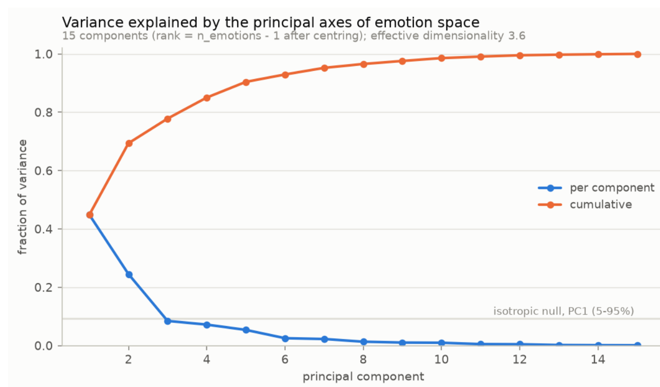
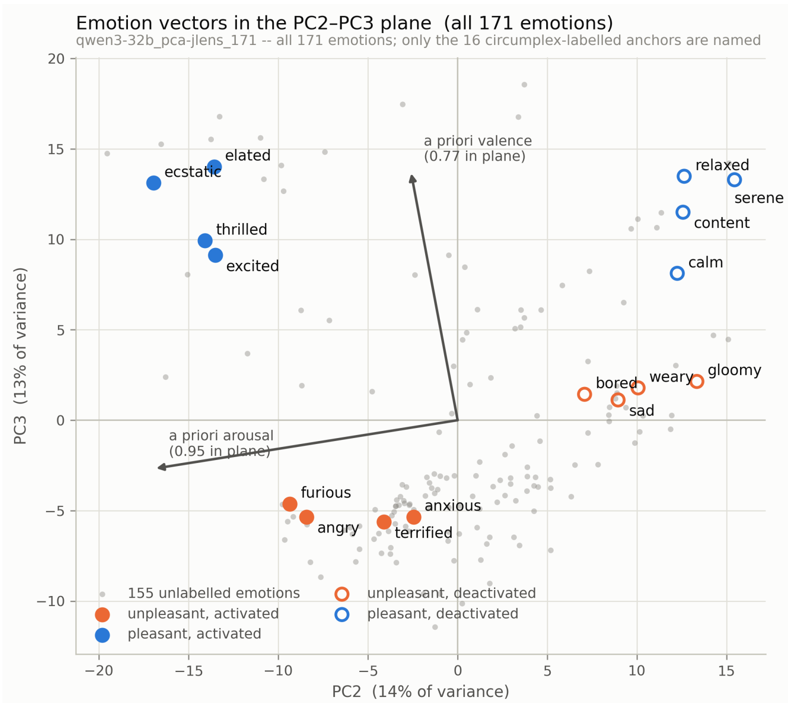
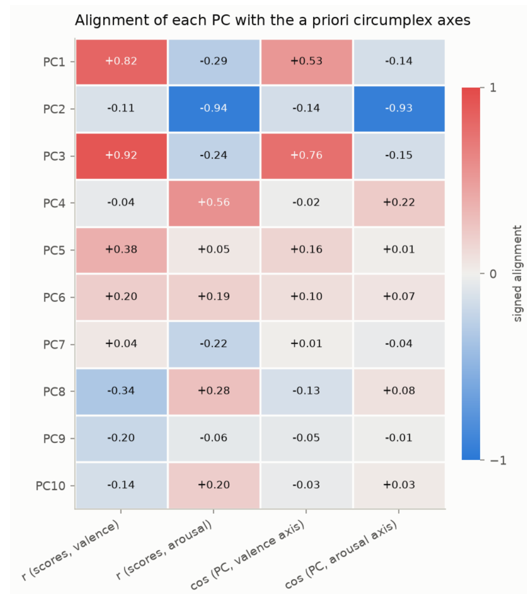


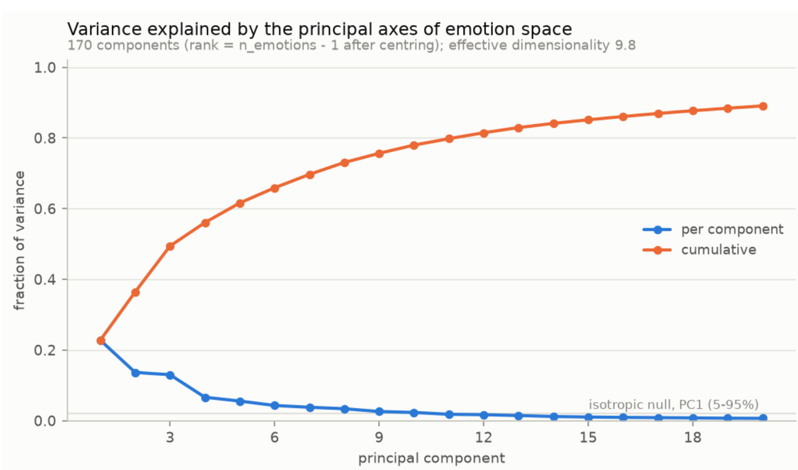
Table 1 -- per-PC J-lens readout, 16-anchor set (compact; for the report body)

	PC	variance %	best axis	ordering AUROC	permutation p	verdict
0	1	44.861029	arousal	0.959184	0.002999	lexicalised
1	2	24.516021	valence	1.000000	0.001499	lexicalised
2	3	8.455365	valence	0.306122	0.369815	murky
3	4	7.200253	arousal	0.469388	0.168916	murky
4	5	5.380553	arousal	0.183673	0.624688	murky

Figures for all 171 emotion vectors







LLM Usage Statement

I used Claude and Codex for feedback on ideas, writing code, and helping to draft writing sections. Results were independently verified.